

The Internet and its usage are constantly evolving. Every single day marks the launch of new and interesting ways of accessing data and interacting with people, pushing the boundaries of HTML in every vertical. The current version of HTML, 4.01, has been in use for almost a decade now. Yet, the publishers are constantly looking out for more evolved techniques to provide enhanced functionalities that have, till now, been restrained by the programming languages as well as the browsers.

To give authors more flexibility and interoperability, and enable more interactive websites and applications, HTML 5 introduces and enhances a wide range of features, including form controls, APIs, multimedia, structure and semantics.

HTML 5 will be the first major change to our lingua franca since XHTML 1.0 in 2000 (the latest being XHTML 2.0, released in 2002). You must have already seen the 'HTML 5 Working Draft' [www.w3.org/TR/html5] at the start of this year. The W3C HTML Working Group and WHATWG (Web Hypertext Application Technology Working Group) have been working extra time, trying to satisfy everyone in an open process. Not an easy task. Sometimes, amongst the concerns and the questions, it's easy to forget that fact.

A lot of us believe that the introduction of the new specifications is just another preposterous attempt by the bigwigs in the browser arena to foist what they want, onto the developers. But then again, there are others who see it as the way forward to develop powerful multimedia Web apps on an open architecture, without Flash, Silverlight or similar proprietary technologies. As Doug Schepers, the W3C's Team Contact for the SVG and Web Apps Working Group says, "HTML 5 is not a technical achievement, it's a social movement."

The varied opinions regarding HTML 5 are because it is much

more than just a mark-up syntax for documents. The very name of the language's specifications working group, Web Hypertext Application Technology Working Group, suggests how much HTML means to the Web. The original goal for HTML 5 was to make it easier to develop Web applications. There's evidence of this in the rash of new JavaScript APIs and support for offline development, some of which are already available in a browser near you.

The elements

So, what is this set of new 'elements' of HTML 5 that's making waves on the Web and elsewhere?

While there is a beautiful article on the Web [www.ibm.com/developerworks/library/x-html5], by Elliott Rusty Harold, that I would strongly recommend you read for a complete overview on the new elements of HTML 5, some of the more important aspects that I plan to discuss in this article are:

- Canvas—inline SVG and MathML
- Audio interface
- Video elements
- Offline Web applications
- Drag and drop
- `getElementsByClassName`
- Web Forms 2.0

However, you can consult Table 1 to figure out which of the elements are already supported by today's browsers.

The latest HTML mark-up

We'll start by thinking about marking up a typical blog. Like the vast majority of sites on the Web, blogs comprise a header, some navigation (often a sidebar or two), a main content area, and a footer.

Currently, there are no ways in HTML 4 to mark up these elements in a semantic fashion—i.e., HTML 4 offers no footer or header elements of its own. Instead, they're usually wrapped in a generic `div` element, a technique that is described in the HTML 4 specification [www.w3.org/TR/html4/struct/global.html#h-7.5.4]: "The DIV and SPAN elements, in conjunction with the `id` and `class` attributes, offer a generic mechanism for adding structure to documents. These elements define content to be inline (SPAN) or block-level (DIV), but impose no other presentational idioms on the content. Thus, authors may use these elements in conjunction with style sheets, the `lang` attribute, etc, to tailor HTML to their own needs and tastes."

When developing the HTML 5

Compatibility structure of HTML 5 elements with today's browsers

	Chrome	Firefox	Internet Explorer	Opera	Safari
<code>contentEditable</code>	Yes	Yes	Yes	Yes	Yes
Stylable Elements (new)	Yes	Yes	No	Yes	Yes
<code>getElementsByClassName</code>	Yes	Yes	No	Yes	Yes
Cross-Documents Messaging	Yes	Yes	Yes	Yes	Yes
Web Forms 2.0	Partial	No	No	Yes	Partial
Drag and Drop	Yes	Yes	Yes	No	Yes
<code><audio></code>	No	3.5	No	No	Yes
<code><video></code>	3.0	3.5	No	Labs release, partial	Yes
<code><canvas></code>	Yes	Yes	No	Yes	Yes
Server-sent DOM Events	No	No	No	Partial	No
Client-side Storage (name/value)	2.x	Yes	Yes	No	Yes

Table 1 (Source: <http://molly.com/html5/html5-0709.html>)